

Question and design

Can sequence-only inference return both observed folds of a fold-switching protein? We ran the exp277 contacts-v1 checkpoint on 67 non-capped pairs. The primary test set has 29 pairs with at least 98% identity, held apart by sequence group from 15 development pairs. Seventeen of those 29 are literally identical. Every method returned at most 16 ranked maps chosen without reference contacts or structures.

The primary comparison is negative

At 100 shared root plus 100 arm rollouts per protein, 10-contact clustered branching improved mean minority-fold enrichment by only 0.013 over independent sampling. The paired 95% interval is -0.014 to +0.043; mean minority recall was essentially unchanged. Blind dual-contact hits were 3/29 versus 2/29. The extra hit failed a 3D check. On the 17 exact-sequence test pairs, both methods had one contact hit and the paired interval still spans zero. We cannot claim a reliable search improvement.

What was actually discovered

For 3j7w/3j7v, both independent and branching top-16 shortlists included maps that reconstructed distinct switching-region structures. With the Fold1 sequence input, region-touch IDDT against (Fold1, Fold2) was (0.477, 0.206) and (0.356, 0.702) for the branching shortlist; the independent shortlist also favored the respective folds. True-contact controls showed comparable separation. This is an existence proof for blind inference discovery, not a branching advantage.

Where the search is limited

Four of seven branch-10 oracle dual-contact hits were present among 200 candidates but missed by the blind 16-map selector. In a separate plain-sampling run, blind dual hits were 2, 3, and 3 at 100, 200, and 500 rollouts; the oracle pool reached 9 by 200. The first priority is better reference-free ranking of coherent minority maps. Contact-level cutoffs are operational and not calibrated as structural thresholds; only four proteins received 3D checks.

What the 500-rollout oracle means

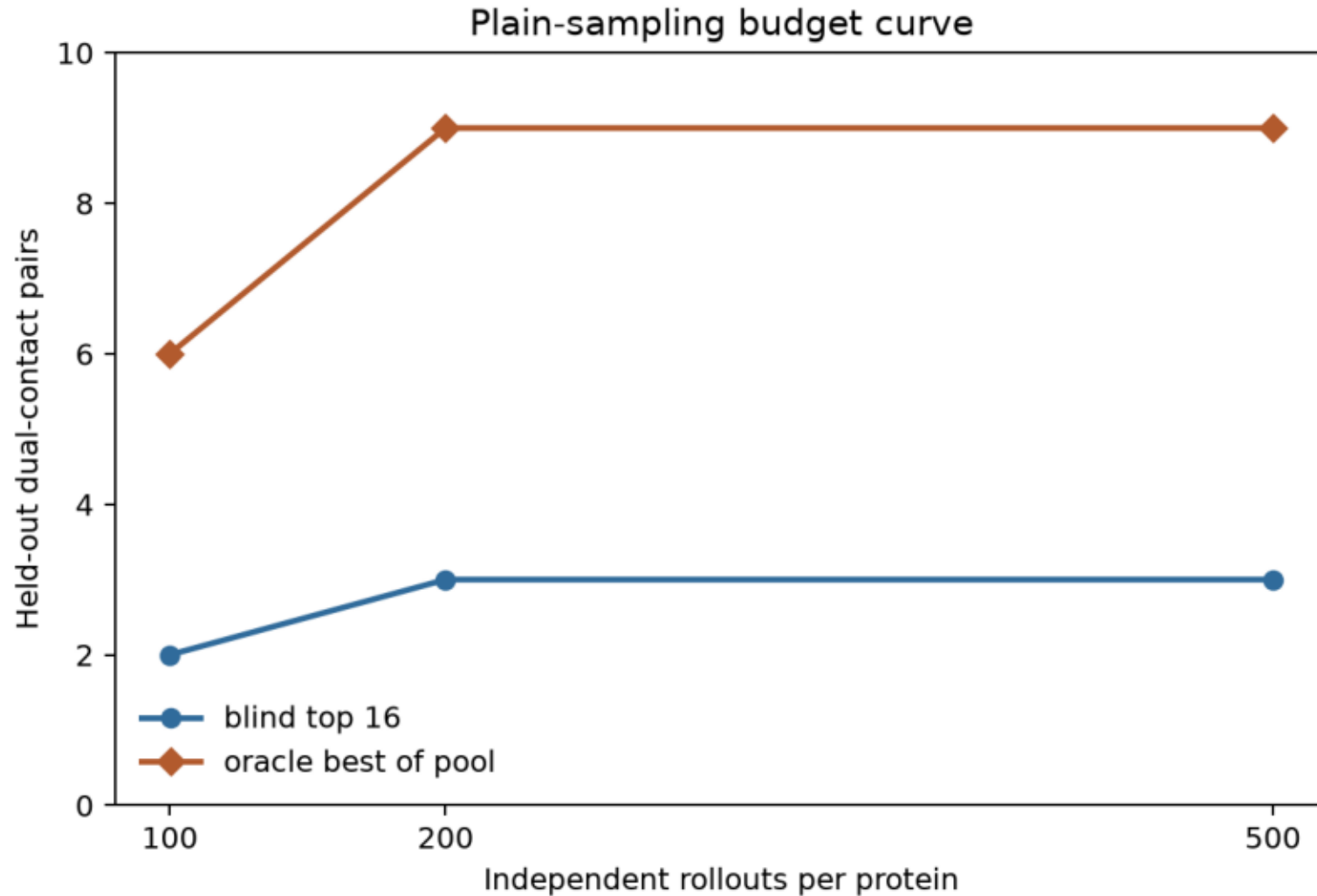
The oracle inspects all 500 independent maps against both reference contact maps. A dual hit needs one Fold1-like and another Fold2-like map; it does not mean the blind selector found either. Among 29 held-out primary pairs, 19 had any Fold1-like map, 13 any Fold2-like map, and 9 had both. The same counts for the 17 exact-sequence test pairs were 12, 7, and 5. Across all 67 non-capped pairs, 31 had both. The prespecified 25% contact-recall screen is permissive: raising recall to 50% reduces held-out dual coverage to 2/29. Only 3j7w/3j7v has a positive 3D validation in this experiment.

Do another 500 iid draws help?

Yes, modestly for oracle contact-pool coverage. On the same 29 primary test proteins, the count with both fold-like contact modes rises from 9 at 500 draws to 11 at 750 and stays 11 at 1,000. Neither-mode proteins fall from 6 to 5. The two new dual hits first appear at draws 509 and 587, and both are exact-sequence pairs. At the stricter 50% recall cutoff, the dual count stays 2/29. These are reference-aware screening hits in the full pool, not blindly selected or newly validated structures. The figure compares the observed stream with averages over random reorderings of these same 1,000 maps; the latter is not an extrapolation.

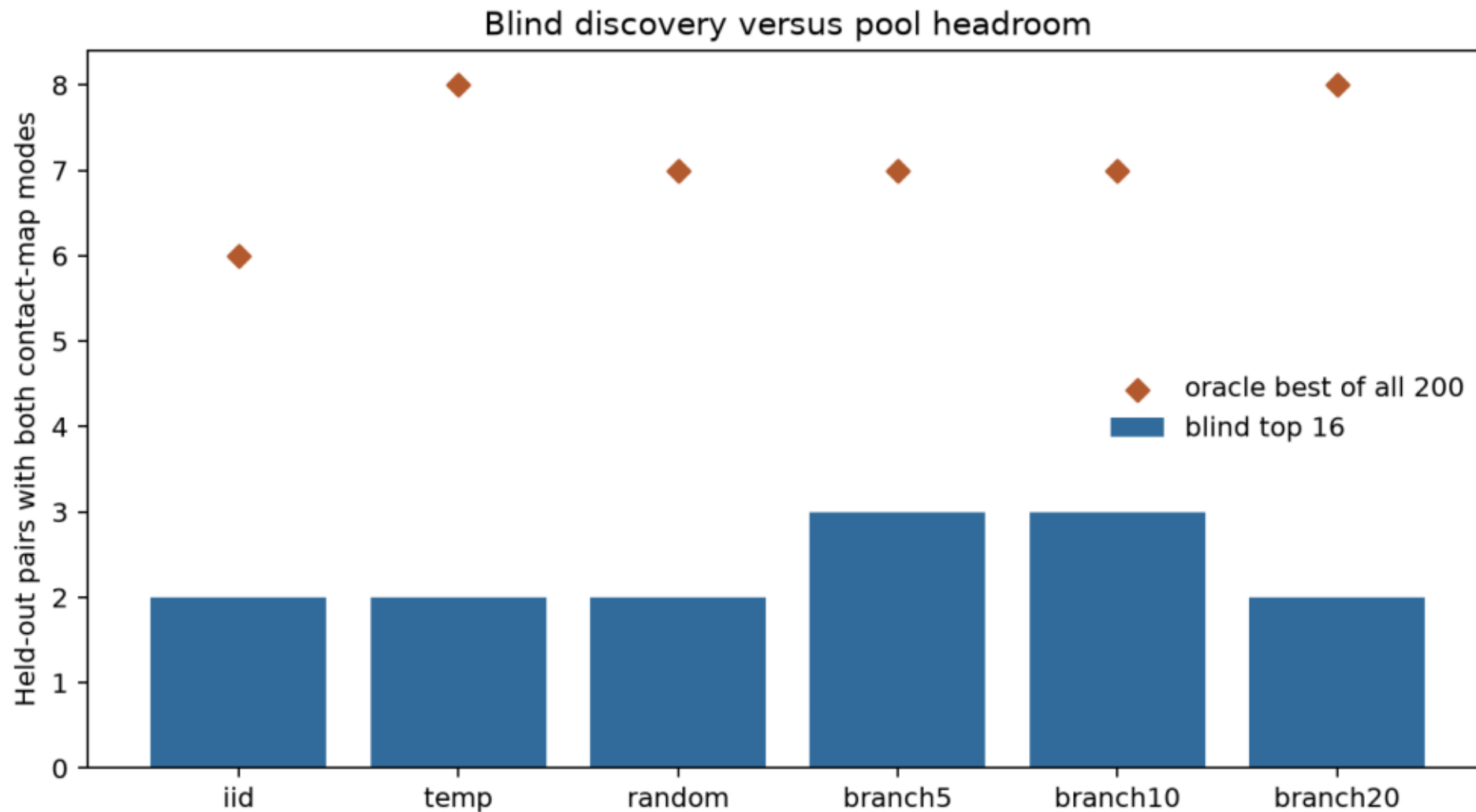
budget_curve.png

A separate 500-rollout run, evaluated at its first 100, 200, and 500 candidates. The blind selector returns 16 maps at each budget; the oracle line is diagnostic and cannot be used at inference time.



dual_contact_hits.png

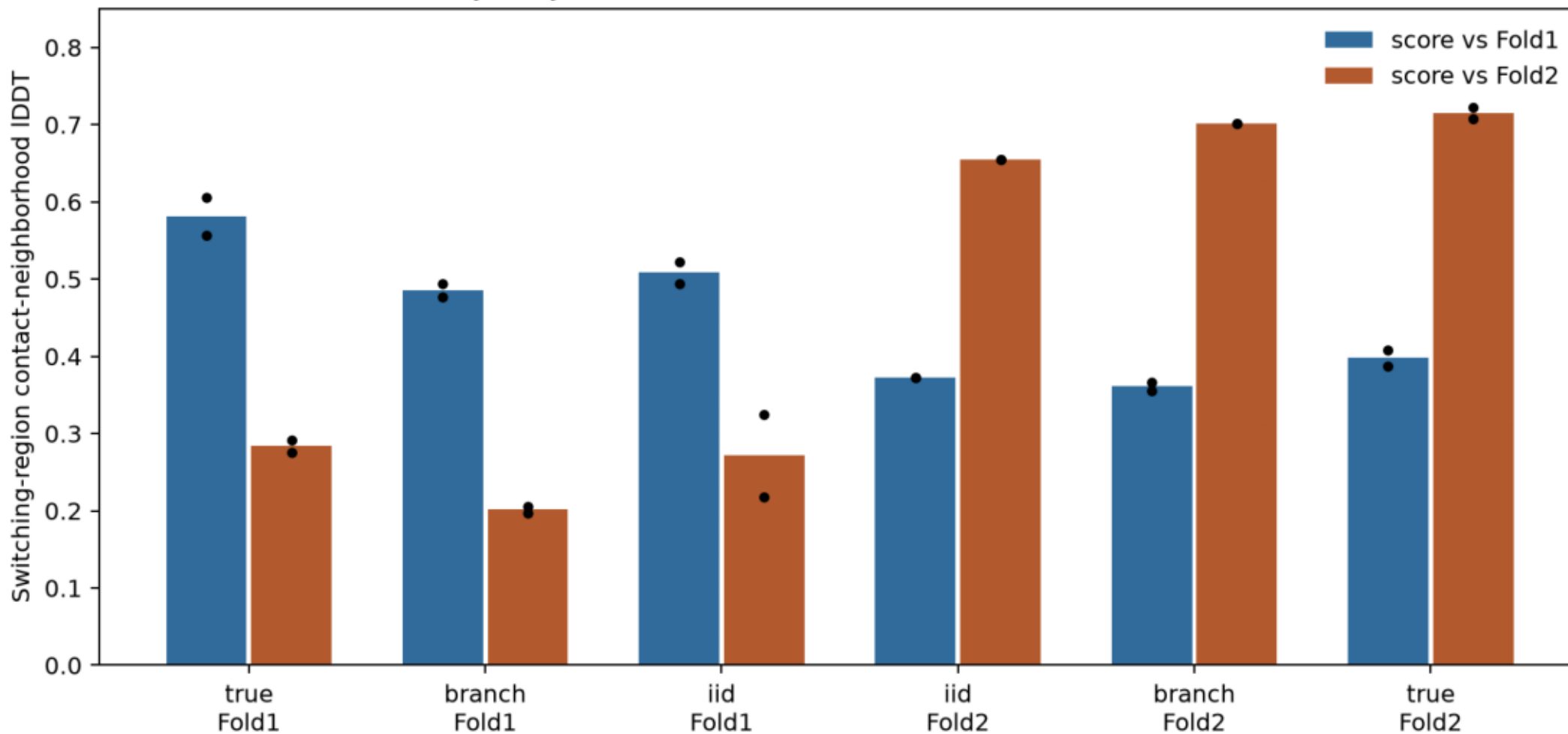
Bars count pairs meeting the prespecified contact-level dual-hit criterion among 16 blindly selected maps. Diamonds show the oracle ceiling in the same candidate pool, which cannot serve as an inference method.



helico_3j7_structural_check.png

Helico structures from two reference input chains; black points are the two inputs. Both 200-rollout shortlist methods yield fold-specific switching-region structures on this protein. True-contact controls show the attainable separation for this reconstruction model.

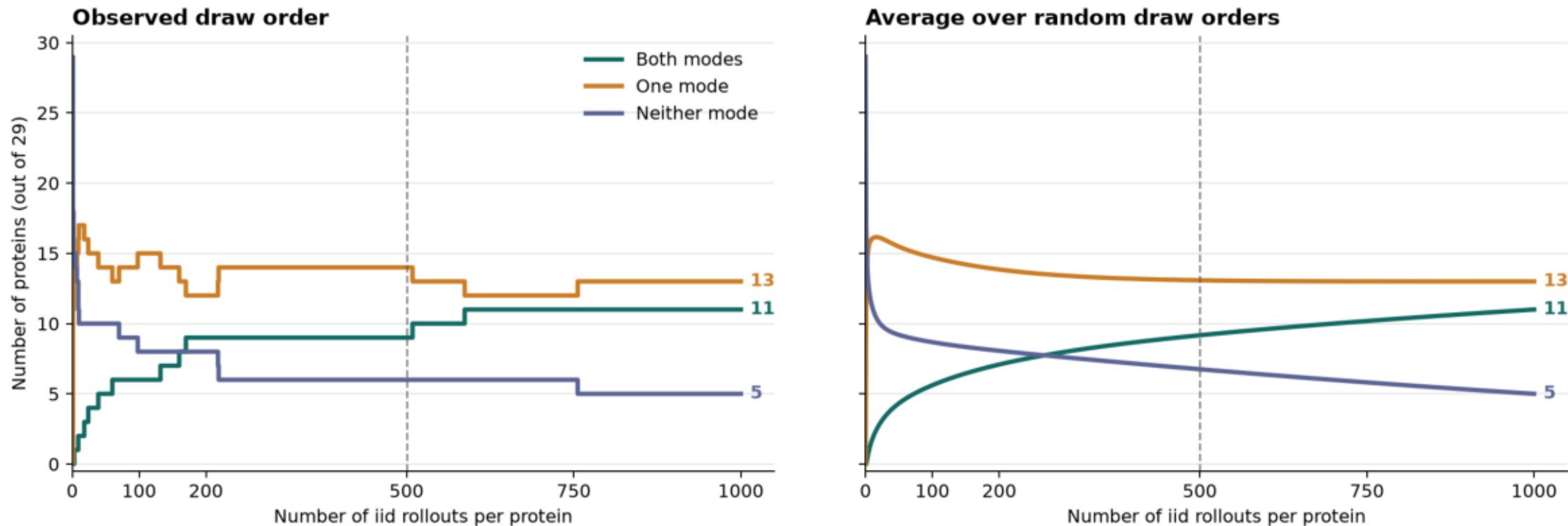
3j7w/3j7v: both blind selectors recover two local folds



iid1000_mode_coverage.png

Cumulative oracle contact-mode coverage among 29 primary test proteins. Left: exact sampled order; right: mean over permutations of the observed 1,000 maps for each protein. Fold-like means a reference-aware contact screening hit, not a verified alternate 3D structure.

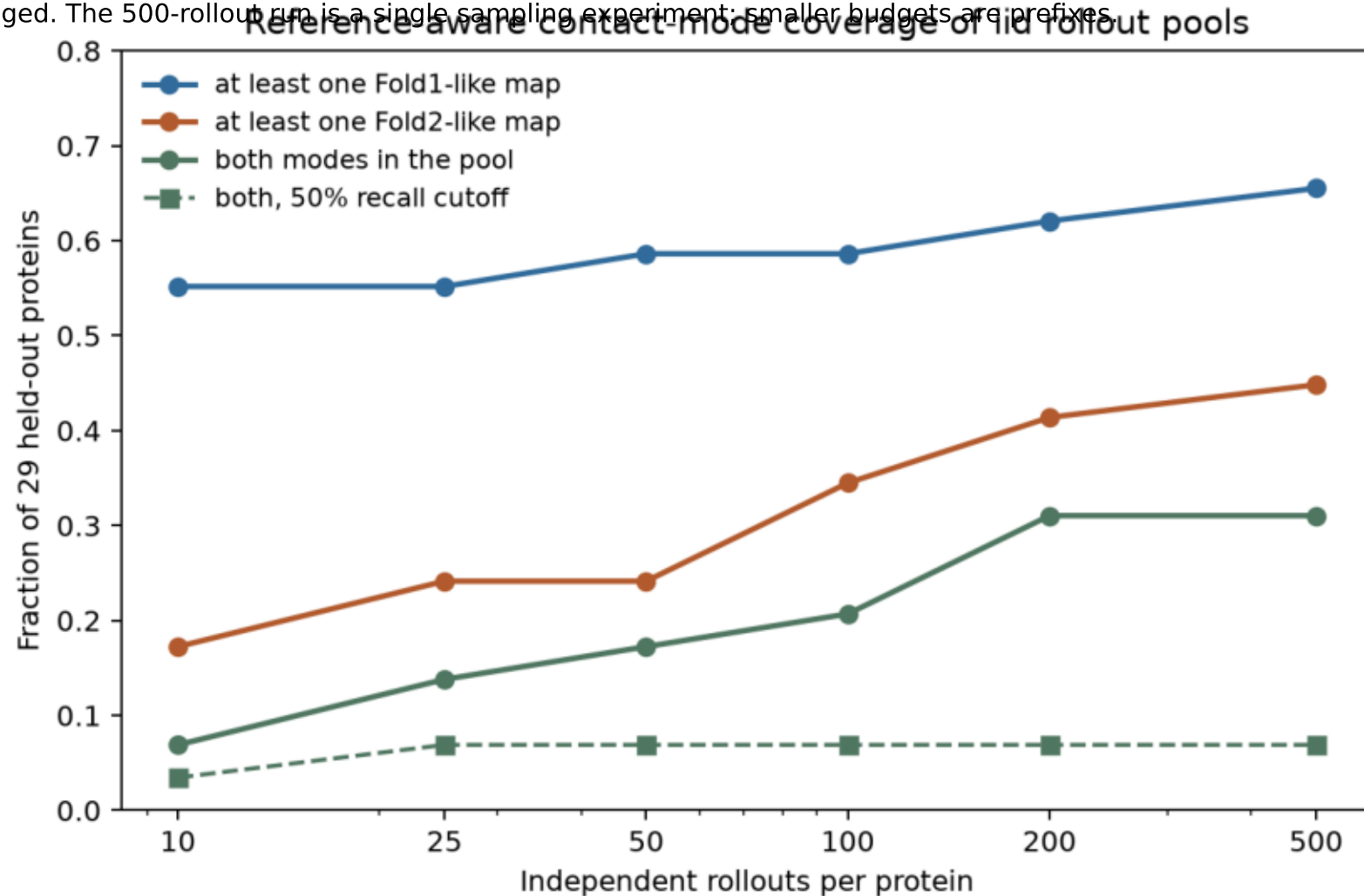
Does more iid sampling reveal both fold-like contact modes?



Each protein is in exactly one category. These are reference-aware contact hits, not verified 3D folds. Dashed line: previous 500-draw limit. Right panel averages permutations of these same 1,000 maps; it is not a forecast beyond 1,000.

iid_mode_coverage.png

All curves inspect every map with reference contacts; none is a blind selection rate. Main hit cutoff: at least 10 switching-region predictions, 25% unique-contact recall, and 0.10 enrichment versus the opposing fold. Dashed curve uses 50% recall with the other conditions unchanged. The 500-rollout run is a single sampling experiment; smaller budgets are prefixes.



paired_enrichment.png

Each point is one held-out primary fold-switching pair. Both methods use the same 100 root rollouts plus 100 arm rollouts; the diagonal is parity.

